

Griffiths Introduction to Quantum Mechanics 3rd
edition Problem 6.1

terU3760

December 2025



Jonah and Nineveh — God commanded Jonah to go to Nineveh. Jonah fled in the opposite direction, yet he was brought back.

In the end, Jonah did go.

Whether he went or refused, God was keeping account all along.

The journey was not forgotten, the delay was not overlooked, and the arrival was fully known by God.

“I went there, and God knew.”

“The journey was not unseen — God bore witness.”

“Not every path needs a human witness; God remembers where I have been.”

“Whether I went or refused, the journey was already known to God.”

(a)

$$\mathbf{r} = (x, y, z), \quad -\mathbf{r} = (-x, -y, -z)$$

$$x = r \sin \theta \cos \phi, \quad y = r \sin \theta \sin \phi, \quad z = r \cos \theta$$

From $\mathbf{r}$ to $-\mathbf{r}$, we only need to:

(i) Reflect on xy -plane:

$$z = r \cos \theta \Rightarrow -z = -r \cos \theta$$

(ii) Rotate: $\phi \rightarrow \phi + \pi$

$$x = r \sin \theta \cos \phi \Rightarrow -x = r \sin \theta \cos(\phi + \pi) = -r \sin \theta \cos \phi$$

$$y = r \sin \theta \sin \phi \Rightarrow -y = r \sin \theta \sin(\phi + \pi) = -r \sin \theta \sin \phi$$

(b)

Similar to (a), since

$$\sin(\pi - \theta) = \sin \theta, \quad \cos(\pi - \theta) = -\cos \theta,$$

$$\sin(\phi + \pi) = -\sin \phi, \quad \cos(\phi + \pi) = -\cos \phi$$

So,

$$\hat{\Pi}\psi(x, y, z) = \hat{\Pi}\psi(r, \theta, \phi) = \psi(r, \pi - \theta, \phi + \pi)$$

$$= \psi(r, r \sin(\pi - \theta) \cos(\phi + \pi), r \sin(\pi - \theta) \sin(\phi + \pi), r \cos(\pi - \theta))$$

$$= \psi(r, -r \sin(\theta) \cos(\phi), -r \sin(\theta) \sin(\phi), -r \cos(\theta))$$

$$= \psi(r, -x, -y, -z)$$

(c)

By Equations (4.15), (4.17), (4.26), (4.22), and (b):

$$\hat{\Pi}\psi_{nlm}(r, \theta, \phi) = \hat{\Pi}R(r)\Theta(\theta)\Phi(\phi)$$

$$= \hat{\Pi}R(r)e^{im\phi}AP_\ell^m(\cos \theta)$$

$$= \hat{\Pi}R(r)e^{im\phi}A(-1)^m(1 - \cos^2 \theta)^{m/2} \left(\frac{d}{d \cos \theta} \right)^m P_\ell(\cos \theta)$$

$$\begin{aligned}
&= \hat{\Pi} R(r) e^{im\phi} A(-1)^m (1 - \cos^2 \theta)^{m/2} \left(\frac{d}{d \cos \theta} \right)^m P_\ell(\cos \theta) \\
&= \hat{\Pi} R(r) e^{im\phi} A(-1)^m (1 - \cos^2 \theta)^{m/2} \left(\frac{d}{d \cos \theta} \right)^m \left[\frac{1}{2^\ell \ell!} \left(\frac{d}{d \cos \theta} \right)^\ell (\cos^2 \theta - 1)^\ell \right] \\
&= R(r) e^{im(\phi+\pi)} A(-1)^m (1 - \cos^2(\pi - \theta))^{m/2} \left(\frac{d}{d \cos(\pi - \theta)} \right)^m \left[\frac{1}{2^\ell \ell!} \left(\frac{d}{d \cos(\pi - \theta)} \right)^\ell (\cos^2(\pi - \theta) - 1)^\ell \right] \\
&= R(r) e^{im\phi} (e^{i\pi})^m A(-1)^m (1 - \cos^2 \theta)^{m/2} (-1)^m \left(\frac{d}{d \cos \theta} \right)^m (-1)^m \left[\frac{1}{2^\ell \ell!} \left(\frac{d}{d \cos \theta} \right)^\ell (-1)^\ell (\cos^2 \theta - 1)^\ell \right] \\
&= (-1)^\ell R(r) e^{im\phi} A(-1)^m (1 - \cos^2 \theta)^{m/2} \left(\frac{d}{d \cos \theta} \right)^m \left[\frac{1}{2^\ell \ell!} \left(\frac{d}{d \cos \theta} \right)^\ell (\cos^2 \theta - 1)^\ell \right] \\
&= (-1)^\ell \psi_{nlm}(r, \theta, \phi)
\end{aligned}$$